

## Percentage

### SSC Practice Sheet - Set 2

#### Percentage Variation

1. A number when reduced by 10% gives 30. The number is  
(1)  $33\frac{1}{2}$  (2)  $33\frac{1}{3}$  ✓  
(3) 40 (4) 35
2. The number of students in a class is increased by 20% and the number now becomes 66. Initially the number was  
(1) 45 (2) 50  
(3) 55 ✓ (4) 60
3. If the monthly salary of an employee is increased by  $2\frac{2}{3}\%$ , he gets 72 rupees more. His monthly salary (in rupees) is  
(1) 7200 (2) 3600  
(3) 2700 ✓ (4) 2000
4. A number increased by  $22\frac{1}{2}\%$  gives 98. The number is  
(1) 45 (2) 18  
(3) 80 ✓ (4) 81
5. A number reduced by 25% becomes 225. What per cent should it be increased so that it becomes 375?  
(1) 25% (2) 30%  
(3) 35% (4) 75%
6. The allowances of an employee constitutes 165% of his basic pay. If he receives Rs. 11925 as gross salary, then his basic pay is (in Rs.)  
(1) 4000 (2) 5000  
(3) 4500 ✓ (4) 5500
7. The numerator of a fraction is increased by 20% and denominator is decreased by 20%. The value of the fraction becomes  $\frac{4}{5}$ . The original fraction is  
(1)  $\frac{2}{3}$  (2)  $\frac{8}{15}$  ✓  
(3)  $\frac{7}{11}$  (4)  $\frac{4}{5}$
8. If the numerator of a fraction is increased by 20% and the

denominator is decreased by 5%, the value of the new fraction becomes  $\frac{5}{2}$ . The original fraction is  
(1)  $\frac{24}{19}$  (2)  $\frac{3}{18}$   
(3)  $\frac{95}{48}$  (4)  $\frac{48}{95}$

9. The sum of two numbers is 520. If the bigger number is decreased by 4% and the smaller number is increased by 12%, then the numbers obtained are equal. The smaller number is;  
(1) 280 (2) 210  
(3) 240 (4) 300

10. The difference between the value of the number increased by 20% and the value of the number decreased by 25% is 36. Find the number.  
(1) 7.2 (2) 0.8  
(3) 720 (4) 80

11. In the expression  $xy^2$ , the values of both variables  $x$  and  $y$  are decreased by 20%. By this, the value of the expression is decreased by  
(1) 40% (2) 80%  
(3) 48.8% (4) 51.2%

#### Percentage Comparison

12. One number is 25% of another number. The larger number is 12 more than the smaller. The larger number is  
(1) 48 (2) 16  
(3) 4 (4) 12

13. If a number  $x$  is 10% less than another number  $y$  and  $y$  is 10% more than 125, then  $x$  is equal to:  
(1) 150 (2) 143  
(3) 140.55 (4) 123.75
14.  $x$  is 5 times longer than  $y$ . The percentage by which  $y$  is less than  $x$  is:  
(1) 50% (2) 40%  
(3) 80% (4) 70%

15. Two numbers are less than a third number by 30% and 37% respectively. How much percent is the second number less than the first?

(1) 10 ✓ (2) 4  
(3) 3 (4) 7

16. The income of C is 20% more than B's and the income of B is 25% more than A's. Find by how much per cent is C's income more than A's?  
(1) 150% (2) 50% ✓  
(3) 25% (4) 35%

17. Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is  
(1) 2 : 5 (2) 3 : 5  
(3) 4 : 5 ✓ (4) 6 : 7

18. First and second numbers are less than a third number by 30% and 37% respectively. The second number is less than the first by  
(1) 7% (2) 4%  
(3) 3% (4) 10%

19. In an examination in which full marks were 500, A got 10% less than B. B got 25% more than C. C got 20% less than D. If A got 360 marks, what percentage of full marks was obtained by D?  
(1) 90% (2) 80% ✓  
(3) 50% (4) 60%

#### Ratio Application

20. If 35% of A's income is equal to 25% of B's income, then the ratio of A's income to B's income is  
(1) 7 : 5 (2) 5 : 7  
(3) 4 : 7 (4) 4 : 3
21. Rani's weight is 25% that of Meena's and 40% that of Tara's. What percentage of Tara's weight is equal to Meena's weight?



- (1) 160% (2) 140%  
(3) 120% (4) 100%

22. If 20% of  $(P + Q) = 50\%$  of  $(P - Q)$ , then find  $P : Q$

- (1) 7 : 8 (2) 7 : 3  
(3) 7 : 5 (4) 5 : 7

23. If 40% of  $(A + B) = 60\%$  of  $(A - B)$  then  $\frac{2A-3B}{A+B}$  is

- (1)  $7/6$  (2)  $6/7$   
(3)  $5/6$  (4)  $6/5$

24. If 120% of  $a$  is equal to 80% of  $b$ , then  $\frac{b+a}{b-a}$  is equal to

- (1) 5 (2) 6  
(3) 7 (4) 8

25. Out of two numbers, 40% of the greater number is equal to 60% of the smaller. If the sum of the numbers is 150, then the greater number is

- (1) 70 (2) 80  
(3) 90 (4) 60

26. Difference of two numbers is 1660. If  $6\frac{1}{2}\%$  of one number is  $8\frac{1}{2}\%$  of the other number, the smaller number is

- (1) 7055 (2) 5395  
(3) 3735 (4) 2075

27. Given that 10% of A's income = 15% of B's income = 20% of C's income. If sum of their income is Rs. 7800, then B's income is:

- (1) 3600 (2) 3000  
(3) 2400 (4) 1800

28. The monthly salaries of A and B together amount to Rs. 40,000. A spends 85% of his salary and B, 95% of his salary. If now their savings are the same, then the salary (in Rs.) of A is

- (1) Rs. 10,000 (2) Rs. 12,000  
(3) Rs. 16,000 (4) Rs. 18,000

29. Two numbers are in the ratio 2 : 3. If 20% of the smaller number added to 20 is equal to the sum of 10% of the larger number and 25, then the smaller number is

- (1) 100 (2) 160  
(3) 180 (4) 200

30. In a class, the number of girls is 20% more than that of the boys. The strength of the class is 66. If 4 more girls are admitted to the class, the ratio of the number of boys to that of the girls is

- (1) 1 : 2 (2) 3 : 4  
(3) 1 : 4 (4) 3 : 5

31. If 20% of  $A = 30\%$  of  $B = \frac{1}{6}$  of  $C$ , then  $A : B : C$  is

- (1) 2 : 3 : 16 (2) 3 : 2 : 16  
(3) 10 : 15 : 18 (4) 15 : 10 : 18

32. If A exceeds B by 60% and B is less than C by 20%, then  $A : C$  is

- (1) 32 : 25 (2) 25 : 32  
(3) 8 : 5 (4) 4 : 5

33. When 60% of a number is subtracted from another number, the second number reduces to its 52%; the ratio of the first number to the second number is :

- (1) 6 : 5 (2) 5 : 3  
(3) 5 : 4 (4) 4 : 5

34. Two numbers A and B are such that the sum of 5% of A and 4% of B is  $\frac{2}{3}$ rd of the sum of 6% of A and 8% of B. The ratio  $A : B$  is

- (1) 4 : 3 (2) 3 : 4  
(3) 1 : 1 (4) 2 : 3

35. The ratio of the number of boys to that of girls in a village is 3 : 2. If 30% of boys and 70% of girls appeared in an examination, the ratio of the number of villagers, appeared in the examination to that not appeared in the same examination is

- (1) 9 : 14 (2) 23 : 27  
(3) 1 : 1 (4) 27 : 23

### Venn Diagrams

36. In an examination, 65% of the students passed in Mathematics, 48% passed in Physics and 30% passed in both. How much per cent

of students failed in both the subjects?

- (1) 17% (2) 43%  
(3) 13% (4) 47%

37. 72% of the students of a certain class took Biology and 44% took Mathematics. If each student took at least one subject from Biology or Mathematics and 40 took both, then the total number of students in the class is :

- (1) 200 (2) 240  
(3) 250 (4) 320

38. In an examination 80% of the boys passed in English and 85% passed in Mathematics, while 75% passed in both. If 45 boys failed in both, the number of boys who sat for the examination was

- (1) 400 (2) 450  
(3) 200 (4) 150

39. In an examination 34% failed in Mathematics and 42% failed in English. If 20% failed in both the subjects, the percentage of students who passed in both subjects was

- (1) 54% (2) 50%  
(3) 44% (4) 56%

40. In a class 60% of the student pass in Hindi and 45% pass in Sanskrit. If 25% of them pass in both the subjects, what percentage of the students fail in both the subjects?

- (1) 80% (2) 75%  
(3) 20% (4) 25%

## Home Work Questions

### Percentage Variation

1. The enhanced salary of a man becomes Rs. 24,000 after 20% increment. His previous salary was

- (1) 20,000 (2) 21,000  
(3) 16,000 (4) 18,000

2. The price of a product is increased by 20%. The new price is Rs. 720, find the initial price of the product.

- (1) Rs. 600 (2) Rs. 700  
(3) Rs. 620 (4) Rs. 580



3. If the numerator of a fraction is increased by 230% and its denominator is increased by 120%, the resulting fraction is  $\frac{9}{4}$ . Find the original fraction.

- (1)  $\frac{11}{8}$  (2)  $\frac{7}{4}$   
(3)  $\frac{13}{10}$  (4)  $\frac{3}{2}$

#### Percentage Comparison

4. A's salary is 50% more than that of B. Then B's salary is less than that of A by

- (1) 50% (2)  $33\frac{1}{3}\%$   
(3)  $33\frac{1}{4}\%$  (4)  $44\frac{1}{2}\%$

5. If the salary of Manoj is 40% less than that of Subhash, then by how much percentage is the salary of Subhash more than that of Manoj?

- (1) 60% (2)  $66\frac{1}{4}\%$   
(3)  $66\frac{2}{3}\%$  (4) 65%

6. Two numbers are respectively  $12\frac{1}{2}\%$  and 25% more than a third number. The first number as percentage of second number is

- (1) 50 (2) 60  
(3) 75 (4) 90

7. Two numbers are less than a third number by 30% and 37% respectively. The per cent by which the second number is less than the first is

- (1) 10% (2) 7%  
(3) 4% (4) 3%

8. A's income is 25% more than B's income. B's income is what per cent of A's income?

- (1) 80 (2) 75  
(3) 50 (4) 25

9. Two numbers are respectively 10% and 25% more than a third number. What per cent is the first of the second?

- (1) 88% (2) 65%  
(3) 75% (4) 80%

10. In an examination A got 25% marks more than B, B got 10% less than C and C got 25% more than D. If

D got 320 marks out of 500, the marks obtained by A were

- (1) 405 (2) 450  
(3) 360 (4) 400

11. X has twice as much money as that of Y and Y has 50% more money than that of Z. If the average money of all of them is Rs. 110, then the money (in Rs.), which X has, is

- (1) 55 (2) 60  
(3) 90 (4) 180

12. p is six times as large as q. The per cent that q is less than p, is

- (1)  $83\frac{1}{3}\%$  (2) 70%  
(3)  $63\frac{1}{3}\%$  (4) 50%

13. Given A is 50% larger than C and B is 25% larger than C, then A is what percent larger than B?

- (1) 25% (2) 50%  
(3) 75% (4) 20%

14. Two numbers are respectively 20% and 50% of a third number. What per cent is the first number of the second?

- (1) 10% (2) 20%  
(3) 30% (4) 40%

15. Two numbers are respectively 25% and 20% less than a third number. What per cent is the first number of the second?

- (1) 5% (2) 75%  
(3) 80% (4) 93.75%

16. If A earns 25% more than B then how much percent does B earn less than A?

- (1) 15% (2) 20%  
(3) 25% (4) 30%

#### Ratio Application

17. If 10% of x is 3 times 15% of y, then find x : y.

- (1) 7 : 2 (2) 9 : 2  
(3) 8 : 3 (4) 11 : 4

18. If 20% of (A + B) = 50% of B, then value of  $\frac{2A-B}{2A+B}$  is

- (1)  $\frac{1}{2}$  (2)  $\frac{1}{3}$   
(3)  $\frac{1}{4}$  (4) 1

19. If 50% of (P - Q) = 30% of (P + Q) and Q = x% of P, then the value of x is:

- (1) 30 (2) 25  
(3) 20 (4) 50

20. If 60% of A's income is equal to 75% of B's income, then B's income is equal to x% of A's income. The value of x is:

- (1) 70 (2) 60  
(3) 80 (4) 90

21. A's salary is 40% of B's salary and B's salary is 25% of C's salary. What percentage of C's salary is A's salary?

- (1) 5% (2) 10%  
(3) 15% (4) 20%

22. 25% of annual salary of A is equal to eighty percent of annual salary of B. Monthly salary of B is 40% of the monthly salary of C. Annual salary of C is Rs. 6 lac. What is the monthly salary of A?

- (1) Rs. 60,000 (2) Rs. 62,000  
(3) Rs. 64,000 (4) Rs. 56,000

23. The difference of two numbers is 45% of their sum. The ratio of the larger number to the smaller number is

- (1) 20 : 9 (2) 9 : 20  
(3) 29 : 11 (4) 11 : 29

24. If 30% of A = 0.25 of B =  $\frac{1}{5}$  of C, then A : B : C is equal to:

- (1) 5 : 6 : 4 (2) 5 : 24 : 5  
(3) 6 : 5 : 4 (4) 10 : 12 : 15

25. The bus fare and train fare of a place from Kolkata were Rs. 20 and Rs. 30 respectively. Train fare has been increased by 20% and the bus fare has been increased by 10%. The ratio of new train fare to new bus fare is

- (1) 11 : 18 (2) 18 : 11  
(3) 5 : 3 (4) 3 : 5

26. The difference of two numbers is 15% of their sum. The ratio of the larger number to the smaller number is

- (1) 23 : 17 (2) 11 : 9  
(3) 17 : 11 (4) 23 : 11



- (1) 160% (2) 140%  
(3) 120% (4) 100%

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(3)  $5/6$  (4)  $6/5$

24. If 120% of  $a$  is equal to 80% of  $b$ , then  $\frac{b+a}{b-a}$  is equal to

- (1) 5 (2) 6  
(3) 7 (4) 8

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